



Flood Risk (Depth-Damage) — Test Results

Automated Test Documentation for Model Governance

MKM Research Labs

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1 Test Results — Flood Risk (Depth-Damage) (MKM-DD-001)

Tests run on **21 March 2026 at 13:30:41**.

Table 1: Test Results Summary — Flood Risk (Depth-Damage) (MKM-DD-001)

Total Tests	128
Passed	128
Failed	0
Skipped	0
Duration	0.01s

Table 2: Detailed Test Results — Flood Risk (Depth-Damage)

Test Description	Acceptance Criteria	Result	Time
At distance = characteristic length, factor must equal $1/e \approx 0.368$.	<code>abs(f - 1.0 / math.e) < 0.001</code>	Pass	0.000s
At 5x the decay length, attenuation should be $\leq 1\%$ ($e^{-5} \approx 0.0067$).	<code>f < 0.01</code>	Pass	0.000s
Negative distance is clamped to zero; must return 1.0.	<code>compute_attenuation(-100) == 1.0</code>	Pass	0.000s
At zero distance, attenuation factor must be 1.0 (no decay).	<code>compute_attenuation(0) == 1.0</code>	Pass	0.000s
Zero decay length means instant decay; must return 0.0 for any distance $\neq 0$.	<code>compute_attenuation(100, 0) == 0.0</code>	Pass	0.000s
Every hour from 0..n must be present in output, in order.	<code>hours == list(range(len(gauge_readings)))</code>	Pass	0.000s
Flood depth must be non-negative at every time step.	<code>r['depth_m'] >= 0.0</code>	Pass	0.000s
Each hydrograph entry must contain hour, wse_m, depth_m, and flooded.	<code>'hour' in r; 'wse_m' in r (+2 more)</code>	Pass	0.000s
Empty gauge readings must return empty property hydrograph.	<code>build_property_hydrograph([], 8.0, 1.0, 0.9, 6.0, 0.3) == []</code>	Pass	0.000s
When floor level exceeds peak WSE, no time step should be flooded.	<code>r['flooded'] is False</code>	Pass	0.000s
With travel time of 5 hours, first 5 hours must show zero flood depth.	<code>r['depth_m'] == 0.0; r['flooded'] is False</code>	Pass	0.000s
Property hydrograph must have same number of time steps as gauge input.	<code>len(result) == len(gauge_readings)</code>	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
Low-elevation property at gauge location must flood at peak water level.	<code>max(peak_depths) > 0</code>	Pass	0.000s
Age 50 to 100	<code>mid_age > young</code>	Pass	0.000s
Age over 100	<code>old > young</code>	Pass	0.000s
Basic returns float	<code>isinstance(score, float); 0 < score <= 10</code>	Pass	0.000s
Construction concrete lower	<code>concrete < brick</code>	Pass	0.000s
Construction steel lowest	<code>steel < concrete</code>	Pass	0.000s
Construction timber higher	<code>timber > brick</code>	Pass	0.000s
Construction wood equals timber	<code>timber == wood</code>	Pass	0.000s
High ltv multiplier	<code>high > low</code>	Pass	0.000s
Ltv 60 to 80 multiplier	<code>mid > base</code>	Pass	0.000s
Ltv 80 to 95 multiplier	<code>mid > base</code>	Pass	0.000s
None ltv no multiplier	<code>0 < score <= 10</code>	Pass	0.000s
Unknown construction type	<code>0 < score <= 10</code>	Pass	0.000s
Unknown flood level	<code>0 < score <= 10</code>	Pass	0.000s
Very high flood capped	<code>score == 10.0</code>	Pass	0.000s
Very low flood lowest score	<code>vl < vh</code>	Pass	0.000s
Slope from 10m to 8m over 1000m should be 0.002 (2m/1000m).	<code>abs(s - 0.002) < 1e-6</code>	Pass	0.000s
Flat terrain (equal elevations) must return MIN_SLOPE, not zero.	<code>s == MIN_SLOPE</code>	Pass	0.000s
Slope must be the same regardless of flow direction (uses absolute value).	<code>s1 == s2</code>	Pass	0.000s
Zero horizontal distance must return MIN_SLOPE to avoid division by zero.	<code>t == 0.0; s == MIN_SLOPE</code>	Pass	0.000s
At zero is max	<code>RiskAssessor.calculate_distance_risk_factor == 1.0</code>	Pass	0.000s
Close band	<code>RiskAssessor.calculate_distance_risk_factor == 0.8</code>	Pass	0.000s
Far band	<code>RiskAssessor.calculate_distance_risk_factor == 0.2</code>	Pass	0.000s
Medium band	<code>RiskAssessor.calculate_distance_risk_factor == 0.4</code>	Pass	0.000s
Monotone decrease	<code>all(factors[i] >= factors[i + 1] for i in range(len(facto...</code>	Pass	0.000s
Near band	<code>RiskAssessor.calculate_distance_risk_factor == 0.6</code>	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
Negative returns default	<code>RiskAssessor.calculate_distance_risk_factor(0)</code> == 0.5	Pass	0.000s
None returns default	<code>RiskAssessor.calculate_distance_risk_factor(0)</code> == 0.5	Pass	0.000s
Very close	<code>RiskAssessor.calculate_distance_risk_factor(5)</code> == 1.0	Pass	0.000s
Very far	<code>RiskAssessor.calculate_distance_risk_factor(50)</code> == 0.1	Pass	0.000s
At high threshold	<code>RiskAssessor.assess_flood_risk_level(1.0)</code> == "High"	Pass	0.000s
At low threshold	<code>RiskAssessor.assess_flood_risk_level(0.1)</code> == "Low"	Pass	0.000s
At medium threshold	<code>RiskAssessor.assess_flood_risk_level(0.5)</code> == "Medium"	Pass	0.000s
Below low threshold	<code>RiskAssessor.assess_flood_risk_level(0.05)</code> == "Low"	Pass	0.000s
High range	<code>RiskAssessor.assess_flood_risk_level(0.75)</code> == "High"	Pass	0.000s
Medium range	<code>RiskAssessor.assess_flood_risk_level(0.3)</code> == "Medium"	Pass	0.000s
Monotone with depth	<code>levels == order</code>	Pass	0.000s
Negative returns unknown	<code>RiskAssessor.assess_flood_risk_level(-1.0)</code> == "Unknown"	Pass	0.000s
None returns unknown	<code>RiskAssessor.assess_flood_risk_level(None)</code> == "Unknown"; <code>RiskAssessor.assess_ltv_risk_level(None)</code> == "Unknown"	Pass	0.000s
Very high	<code>RiskAssessor.assess_flood_risk_level(2.5)</code> == "Very High"	Pass	0.000s
Zero is very low	<code>RiskAssessor.assess_flood_risk_level(0.0)</code> == "Very Low"	Pass	0.000s
Daily minus 15	<code>r['reliability_score'] == 80</code>	Pass	0.000s
Decommissioned	<code>r['reliability_score'] == 0;</code> <code>r['category'] == "Unreliable"</code>	Pass	0.000s
Fully operational	<code>r['reliability_score'] == 95;</code> <code>r['category'] == "Highly Reliable"</code>	Pass	0.000s
Hourly minus 5	<code>r['reliability_score'] == 90</code>	Pass	0.000s
Maintenance required	<code>r['reliability_score'] == 70;</code> <code>r['category'] == "Reliable"</code>	Pass	0.000s
Monthly minus 50	<code>r['reliability_score'] == 20;</code> <code>r['category'] == "Unreliable"</code>	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
Realtime no adjustment	<code>rt['reliability_score'] ==</code> <code>base['reliability_score']</code>	Pass	0.000s
Result keys	<code>'reliability_score' in r;</code> <code>'category'</code> <code>in r (+1 more)</code>	Pass	0.000s
Score clamped at zero	<code>r['reliability_score'] == 0</code>	Pass	0.000s
Temporarily offline	<code>r['reliability_score'] == 30;</code> <code>r['category'] == "Low Reliability"</code>	Pass	0.000s
Unknown status	<code>r['reliability_score'] == 50;</code> <code>r['category'] == "Moderately</code> <code>Reliable"</code>	Pass	0.000s
Weekly minus 30	<code>r['reliability_score'] == 65</code>	Pass	0.000s
All items are strings	<code>all(isinstance(r, str) for r in</code> <code>recs)</code>	Pass	0.000s
Deep flood extra items	<code>len(deep) > len(shallow)</code>	Pass	0.000s
High risk four or more	<code>len(recs) >= 4</code>	Pass	0.000s
High risk mentions flood insurance	<code>any("flood insurance" in r.lower()</code> <code>for r in recs)</code>	Pass	0.000s
Low risk	<code>len(recs) >= 2</code>	Pass	0.000s
Medium risk	<code>len(recs) >= 3</code>	Pass	0.000s
Very low risk	<code>len(recs) >= 1</code>	Pass	0.000s
Base rates by level	<code>result == pytest.approx(rate)</code>	Pass	0.000s
Capped at 10 pct	<code>result == pytest.approx(0.10)</code>	Pass	0.000s
Depth increases premium	<code>with_depth > no_depth</code>	Pass	0.000s
Depth multiplier formula	<code>rate == pytest.approx(expected)</code>	Pass	0.000s
Unknown level uses default	<code>result == pytest.approx(0.005)</code>	Pass	0.000s
Zero depth no effect	<code>base == zero</code>	Pass	0.000s
At high boundary	<code>RiskAssessor.assess_ltv_risk_level(0.95)</code> <code>== "High"</code>	Pass	0.000s
At low boundary	<code>RiskAssessor.assess_ltv_risk_level(0.6)</code> <code>== "Low"</code>	Pass	0.000s
At moderate boundary	<code>RiskAssessor.assess_ltv_risk_level(0.8)</code> <code>== "Moderate"</code>	Pass	0.000s
Critical ltv	<code>RiskAssessor.assess_ltv_risk_level(0.98)</code> <code>== "Critical"</code>	Pass	0.000s
High ltv	<code>RiskAssessor.assess_ltv_risk_level(0.88)</code> <code>== "High"</code>	Pass	0.000s
Low ltv	<code>RiskAssessor.assess_ltv_risk_level(0.5)</code> <code>== "Low"</code>	Pass	0.000s
Moderate ltv	<code>RiskAssessor.assess_ltv_risk_level(0.7)</code> <code>== "Moderate"</code>	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
None returns unknown	<code>RiskAssessor.assess_flood_risk_level(None)</code> == "Unknown"; <code>RiskAssessor.assess_ltv_risk_level(None)</code> == "Unknown"	Pass	0.000s
Normalises 95 pct	<code>RiskAssessor.assess_ltv_risk_level(95)</code> == "High"	Pass	0.000s
Normalises percentage input	<code>RiskAssessor.assess_ltv_risk_level(70)</code> == "Moderate"	Pass	0.000s
Monotonicity: deeper water must produce higher velocity at constant slope.	<code>v2 > v1</code>	Pass	0.000s
Negative depth is physically invalid; must return zero velocity.	<code>compute_manning_velocity(-1.0, 0.005)</code> == 0.0	Pass	0.000s
Negative slope (downhill in opposite direction) must use absolute value.	<code>v_pos == v_neg</code>	Pass	0.000s
Manning velocity at d=1.0m, S=0.005, n=0.04 should be ~1.77 m/s.	<code>1.5 < v < 2.0</code>	Pass	0.000s
Slopes below MIN_SLOPE must be clamped to MIN_SLOPE to avoid	<code>v == v_min</code>	Pass	0.000s
Monotonicity: steeper slope must produce higher velocity at constant depth.	<code>v2 > v1</code>	Pass	0.000s
Zero depth must produce zero velocity (no flow without water).	<code>compute_manning_velocity(0.0, 0.005)</code> == 0.0	Pass	0.000s
Zero roughness coefficient causes division by zero; must return 0.	<code>compute_manning_velocity(1.0, 0.005,</code> <code>roughness=0.0) == 0.0</code>	Pass	0.000s
Elevated ltv medium flood	"Moderate Risk" in r	Pass	0.000s
High flood overrides	"High Risk" in r and "flood" in <code>r.lower()</code>	Pass	0.000s
High ltv medium flood	"High Risk" in r	Pass	0.000s
Low flood risk	"Low Risk" in r	Pass	0.000s
Medium flood default	"Moderate Risk" in r	Pass	0.000s
Minimal risk	"Minimal" in r	Pass	0.000s
Negative value critical	"Critical Risk" in r	Pass	0.000s
Negative value high	"High Risk" in r	Pass	0.000s
Negative value moderate	"Moderate Risk" in r	Pass	0.000s
None ltv no crash	<code>isinstance(r, str)</code>	Pass	0.000s
Very high flood overrides	"High Risk" in r	Pass	0.000s
Close to water increases score	<code>near['vulnerability_score'] ></code> <code>far['vulnerability_score']</code>	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
Flooded depth computed	<code>r['flood.depth'] == pytest.approx(1.0); r['vulnerability.score'] > 0</code>	Pass	0.000s
High risk recommendations present	<code>any("flood" in rec.lower() for rec in r['recommendations'])</code>	Pass	0.000s
No flood	<code>r['flood.depth'] == 0.0; r['vulnerability.score'] == 0</code>	Pass	0.000s
None elevation returns unknown	<code>r['risk.level'] == 'Unknown'; r['flood.depth'] is None (+1 more)</code>	Pass	0.000s
None flood level returns unknown	<code>r['risk.level'] == 'Unknown'</code>	Pass	0.000s
Result dict keys	<code>set(r.keys()) == {'flood.depth', 'risk.level', 'vulnerabi...</code>	Pass	0.000s
Score capped at 100	<code>r['vulnerability.score'] <= 100</code>	Pass	0.000s
Travel time for 500m at d=1.0m, S=0.005 should be well under 1 hour.	<code>0 < t < 1.0</code>	Pass	0.000s
Monotonicity: greater distance must produce longer travel time.	<code>t2 > t1</code>	Pass	0.000s
Zero depth produces zero velocity, so travel time must be infinite.	<code>t == float('inf')</code>	Pass	0.000s
Zero distance must produce zero travel time.	<code>t == 0.0; s == MIN_SLOPE</code>	Pass	0.000s
Base pct wins when higher than depth factor	<code>var == pytest.approx(350.000)</code>	Pass	0.000s
Depth 1 to 2m	<code>var == pytest.approx(500.000)</code>	Pass	0.000s
Depth half to 1m	<code>var == pytest.approx(250.000)</code>	Pass	0.000s
Depth over 2m	<code>var == pytest.approx(800.000)</code>	Pass	0.001s
Depth under half m	<code>var == pytest.approx(100.000)</code>	Pass	0.000s
Negative property value	<code>RiskAssessor.calculate_value_at_risk(-100.000, "High") ==...</code>	Pass	0.000s
Risk level percentages	<code>var == pytest.approx(1.000.000 * pct)</code>	Pass	0.000s
Zero depth uses base pct	<code>var == pytest.approx(150.000)</code>	Pass	0.000s
Zero property value	<code>RiskAssessor.calculate_value_at_risk(0, "High") == 0.0</code>	Pass	0.000s

1.1 Test Document History

Date	Version	Description	Author
05-Mar-2026	1.0	Initial test suite (29 tests)	D. Kelly
07-Mar-2026	1.1	29 test(s) added; 29 test(s) removed (total: 29)	D. Kelly
18-Mar-2026	1.2	117 test(s) added (total: 146)	D. Kelly
21-Mar-2026	1.3	18 test(s) removed (total: 128)	D. Kelly